To: The Centers for Medicare & Medicaid Services (CMS) and the Assistance Secretary for Technology Policy (ASTP)

From: Keith W. Boone, Patient, Health IT Standards Guru

Date: June 15, 2025

RE: Request for Information; Health Technology Ecosystem (CMS-042-NC)

Introduction

I am responding to this RFI not as an employee of a health information technology vendor, but rather as a well informed patient and caregiver for my wife, children and mother. My feedback is based upon my own personal interactions with Health IT systems, and are informed by my experience of a developer of multiple Health Information Network technologies, certified EHR systems, and contributing author to many of the standards upon which these systems are based.

I have two adult children living at home, my daughter who recently completed her associates degree and is currently employed part time, and my son, who cares for disabled partner who also lives with us. My wife is retired. My mother lives more than a thousand miles away, but my wife and I (and my brother and his spouse) provide care and assistance for her through several health conditions that have occurred in her senior years, including a heart attack, and bowel cancer.

As a general comment and in the interest of regulatory simplicity:

Many of the questions in this RFI are repetitive, asked from the perspective of patients, providers, payers, networks and technology vendors. I have not responded to each section. In the future, CMS might ask one question about what perspective the respondent identifies with, and then just ask all of the questions in one place, and sort the responses individually as necessary. You are going to get provider perspectives in the payer section, why ask the same question twice in each when you have to sort it out anyway.

Thank you for giving me the opportunity to comment on these issues.

Keith W. Boone

Question Responses

1. Patient Needs

PC-1. What health management or care navigation apps would help you understand and manage your (or your loved ones) health needs, as well as the actions you should take?

In 2016 I moved, and so had to choose a new care provider. I specifically chose a care provider that used Epic, so that I could use MyChart. It wasn't that I was a fan of Epic, in fact, I worked for a competitor. However, I wanted the best capabilities in the patient portal that I could find. While MyChart mostly works for me, it's essentially only the best of a bad lot. I'm frustrated as a patient that I don't have a choice of applications available to me to manage my care independently of the practice that I use.

a. What are the top things you would like to be able to do for your or your loved ones' health that can be enabled by digital health products?

For myself, and for each of my family members who have authorized me I want to be able to schedule and reschedule appointments, ask questions of the care team or specific care team members, see my medications and refill prescriptions, look at lab results as individual values, or trended over time, see my immunization history as known to my state of residence, and what my provider knows, and see new diagnoses. I also want to be able to see my images and other test results. I might even want to pay my bills. With the exception of imaging, and paying my bills, I have been able to do most of these using MyChart, but no other patient facing application seems to be available to me. I cannot view everything as a consolidated record of care, only through separate MyChart accounts with my primary care provider, specialists, the two hospitals we generally use, my pharmacy and my payers.

While I can do most of this this with MyChart, I have yet to see a unified view across all my access through MyChart. I also get data via Quest and LabCorp, some routed to my iPhone through their respective portals. While accessible through my personal device, it's not integrated with my view through MyChart. I don't have any real access through Payer data, and most payer portals I have ever used are a world of complexity, poor user interface design, and not really designed for any sort of management of my healthcare or its finances.

b. If you had a personal assistant to support your health needs, what are the top things you would ask them to help with? In your response, please consider tasks that could be supported or facilitated by software solutions in the future.

Siri/Alexa/Google/ChatGpt (whoever, whatever):

- please ask my doctor to renew my ____. .
- Please ask my son's doctor to renew his ___.
- Please schedule an ob/gyn visit for my daughter.
- What is my last HgBA1C? How has it been trending over the last year?
- How many steps did I walk last week?

- What's the lowest cost place for ____ to get a (diagnostic or imaging test)?

PC-2. Do you have easy access to your own and all your loved ones' health information in one location (for example, in a single patient portal or another software system)?

As previously stated, mostly I use MyChart. With that, I can access hospital, primary care, specialist and pharmacy data, as well as labs. Easy? Not really. I have multiple providers so it is not all in one unified location or login, I also have my lab data available via Quest and LabCorp through their portals.

a. If so, what are some examples of benefits it has provided?

I'm often able to help my wife with her health care issues, scheduling visits and asking providers questions on her behalf that she doesn't know how to word. I can do the same for myself.

b. If not, in what contexts or for what workflows would it be most valuable to use one portal or system to access all such health information?

Today I have 6 different logins to access all of my health data, and absolutely NO single application to access it. Oh yes, I also have data available from my payers (both past and current), and two different sites for HSA or FSA data but I hardly ever use it as it is next to useless, hard to access, and just ONE MORE login. Finally, I separately have a portal to track my FSA and HSA funds, which is separate from any of the above. So, 8-10 logins total, and it's a royal pain to find the right username and password for each.

c. Were there particular data types, such as x-rays or specific test results, that were unavailable? What are the obstacles to accessing your own or your loved ones' complete health information electronically and using it for managing health conditions or finding the best care (for example, limitations in functionality, user friendliness, or access to basic technology infrastructure)?

I cannot get to imaging results unless I ask for access from the imaging provider, and then it's a CD, with its own viewer app, which is today, worse than a faxed image.

PC-3. Are you aware of health management, care navigation, or personal health record apps that would be useful to Medicare beneficiaries and their caregivers?

Not yet, as I'm NOT in that space. My employer will happily provide me with such an application, and electronic devices to track my weight, and apps for diet, et cetera. I find these generally unhelpful as they have limited data access.

PC-4. What features are missing from apps you use or that you are aware of today?

The biggest missing capability is apps that really are designed for patient use, which can access data from multiple providers and payers and aggregate it across them, instead of adjuncts such as MyChart which only work with vendor specific EHR systems.

a. What apps should exist but do not yet? Why do you believe they do not exist yet?

I believe that they do not exist yet, because it is very difficult to extract value from the users of them by the companies that would develop such an application. Health data is very private, and very protected. Many patients (including me), wouldn't want to offer up our data as the value proposition, and few would be willing to pay enough to make such an application profitable. I pay huge $ to my insurance company, but I wouldn't trust them to provide an app, as their own portals generally suck. I'd rather have the ability to have my provider "prescribe an app" which I could choose, and have my insurance pay for it.

Another capability I expect would be missing would be aggregation across the streams. Apps seem to have paid too much attention to GhostBusters, and never cross the streams from multiple sources. Imagine them trying to deal with immunization data from the state registry and my healthcare provider, or multiple instances of the same diagnosis reported by multiple providers, or the lab result my provider reports and the one that I get from LabCorp that is in fact the same lab result.

b. What set of workflows do you believe CMS is uniquely positioned to offer?

Offer application developers the ability to be paid by CMS (or other payers), and payers the capacity to pay for one healthcare application of the patients choosing which meets a minimum set of functional criteria.

Enable patients to share access to that application with authorized family members who facilitate healthcare for them. If my wife's insurer pays for an application for her, that singular fee should enable my wife and any non-provider care giver she authorizes to use that application on as many devices as desired (she has a computer, tablet and mobile phone, I, her assistant in these activities, have 3 tablets, 2 computers, and 2 phones).

Also, enable patients to use as many applications as they want to access their data from EHRs, payers, and other sources, and don't make it cost them anything to connect to those endpoints with the data.

Make it possible for patients to establish a single credential which can be used to enable application access from any provider or payer portal.

PC-5. What can CMS and its partners do to encourage patient and caregiver interest in these digital health products?

CMS can encourage patient interest by creating a useful market for these digital health products that is PATIENT facing, not provider, payer, or pharma facing. A patient facing market is one which gives the patient a choice of applications that serve their needs. Present "patient portal" markets are mostly provider or payer facing. They "have to be available", but in fact, they aren't really patient oriented or focused.

a. What role, if any, should CMS have in reviewing or approving digital health products on the basis of their efficacy, quality or impact or both on health outcomes (not approving in the sense of a coverage determination)? What criteria should be used if there is a review process? What

technology solutions, policy changes, or program design changes can increase patient and caregiver adoption of digital health products (for example, enhancements to data access, reimbursement adjustments, or new beneficiary communications)?

CMS should designate minimal functional requirements that an application must have, which should be at least equivalent to “patient portal” capabilities found in ASTP’s Certification requirements. A certifier should approve such patient applications, similar to how EHR systems are today certified (if not in fact using the exact same process for certification criteria).

Reimbursement adjustments should be enough to make payers willing to provide such applications to their patients, but no so much as to make them greedy enough to try to dominate the space with their own applications nor enable them to give their own applications any preferential treatment. In fact, it should almost be verboten for payers to provide patients with their own applications to meet the goal being proposed.

b. What changes would enable timely access to high quality CMS and provider generated data on patients?

Enabling linking of payment data that CMS has back to clinical data that providers have within the patient facing application.

PC-6. What features are most important to make digital health products accessible and easy to use for Medicare beneficiaries and caregivers, particularly those with limited prior experience using digital tools and services?

That’s a usability question. I’m an advanced technology user; you don’t want to ask me that question 😉. Realistically, the most important “feature” that I can think of is usability testing, as is promoted for certified EHR systems.

PC-7. If CMS were to collect real-world data on digital health products' impact on health outcomes and related costs once they are released into the market, what would be the best means of doing so?

1. Ensure outcomes can be determined from the data available within the application. Don’t define measures that need physician intervention to be evaluated wrt to outcome.
2. Tie payment costs to outcomes in traceable ways.

2. Data Access and Integration

PC-8. In your experience, what health data is readily available and valuable to patients or their caregivers or both?

a. What data is valuable, but hard for patients and caregivers, or app developers and other technical vendors, to access for appropriate and valuable use (for example, claims data, clinical data, encounter notes, operative reports, appointment schedules, prices)?

As previously mentioned, my imaging data is damn difficult to get to. Also, while I have all my claims, and my pharmacy spending data, I have no clue what my annual costs are for each condition that could be tied back to those claims or charges. There's no way presently to tie adverse health outcome costs (e.g., preventable hospital visits) back to existing conditions either.

b. What are specific sources, other than claims and clinical data, that would be of highest value, and why?
Likely any data you can get on lifestyle (e.g., exercise and diet) or environment.

c. What specific opportunities and challenges exist to improve accessibility, interoperability and integration of clinical data from different sources to enable more meaningful clinical research and generation of actionable evidence?
Research is useful, but don't overscope the patient application access problem looking to boil the ocean. Look at research as something to address in the future. Let programs like All Of Us cover that aspect for now.

PC-9. Given that the Blue Button 2.0 API only includes basic patient demographic, Medicare coverage, and claims data (Part A, B, D), what additional CMS data sources do developers view as most valuable for inclusion in the API to enable more useful digital products for patients and caretakers?

Link actual coverage dollars to conditions in the data provided.

a. What difficulties are there in accessing or utilizing these data sources today?

b. What suggestions do you have to improve the Blue Button 2.0 API experience?

c. Is there non-CMS data that should be included in the API?

Provide incentives to enable providers to link clinical data in some way to claim data. For example, provide applications a way to get to clinical data elements relevant to a claim through G(10) providers of certified EHR systems based on a claim or EOB identifier. A claim represents in some way, e.g., through a FHIR EpisodeOfCare resource.

[base]/EpisodeOfCare?identifier=[claimid] would allow an EHR to return information for a claim. [base]/EpisodeOfCare/{id}/$everything would allow it to return everything associated with that episode. Enforce support of the FHIR episodeOfCare extension on USCDI elements accessible through g(10) capabilities of certified EHR systems.

PC-10. How is the Trusted Exchange Framework and Common Agreement™ (TEFCA™) currently helping to advance patient access to health information in the real world?

In 2025, TEFCA is still an emerging space. Let it continue to emerge.

a. Please provide specific examples.
Recently EPIC announced that more than 1000 facilities will have access to data through TEFCA. I know of at least one dominant EHR system for patients in LTPAC settings can access data via TEFCA.

b. What changes would you suggest?

Don't just enable, but require participants in TEFCA to respond to public health inquiries.

c. What use cases could have a significant impact if implemented through TEFCA?

Consider cases in public health where there can be a place for public health to offer data via TEFCA, e.g., immunization records, vital records, et cetera, for providers, patients or other public health agencies to be able to access and use the data to improve the health of individuals and populations.

d. What standards are you aware of that are currently working well to advance access and existing exchange purposes?

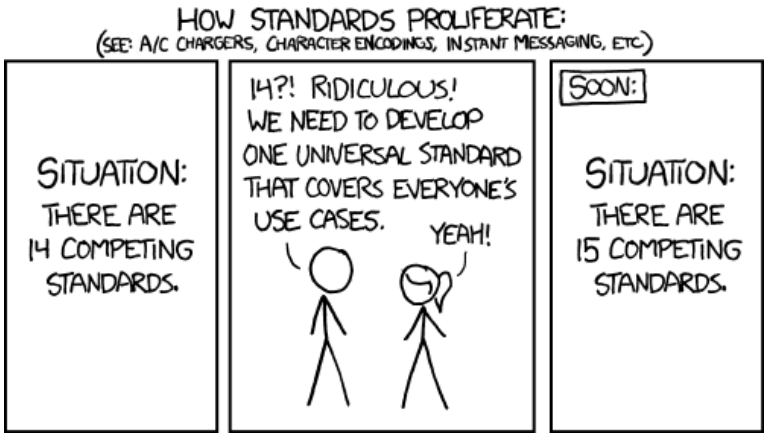
FHIR R4, FAST, OpenID, SMART

e. What standards are you aware of that are not currently in wide use, but could improve data access and integration?

See previous comment about scope, although for imaging, require some use of DICOM or more specifically DICOM Web capabilities would be helpful.

f. Are there redundant standards, protocols, or channels that should be consolidated?

There are always redundant standards. For a simple example, the QHIN Technical Framework (QTF) R1 is very different from QTF R2, yet they provide very similar data access, just at different levels of granularity. Don't try to consolidate standards. That will just lead to this:



g. Are there adequate alternatives outside of TEFCA for achieving widespread patient access to their health information?

Today, with the national networks, we have some inter-network connectivity, BUT it's not as readily available as what would be available through TEFCA. It's as if we let SPRINT, Verizon, AT&T, Mint, and other mobile health providers define their own standards for mobile telephones, but no mandated interoperability between them, so that if I had an AT&T enabled phone, I could only call other parties who also had an AT&T enabled phone.

PC-11. How are health information exchanges (HIEs) currently helping to advance patient access to health information in the real world?

a. How valuable, available, and accurate do you find the data they share to be?

I live in Massachusetts. Sorry, my state doesn't serve me personally all that well here (although it seems to do so for payers and to some degree providers). Others do. The accuracy of the data available in the content is NOT something typically controlled (although it may be monitored) by the HIE, because the data sources are EHR and other Health IT systems, and so the quality is only as good as the connected information providers. Some EHRs I trust more than others, and these days, I actually choose my provider (when I can) based on the EHR system they use.

b. What changes would you suggest?

Mandated abilities for patients and their designated agents to readily access HIE data via the same protocols and standards required by EHRs 42 CFR 170.315(g)(10), and a patient portal providing access to clinical documents encoded using standards in 42 CFR 170.315(b)(1).

c. Are there particular examples of high-performing HIE models that you believe should be propagated across markets?

d. What is the ongoing role of HIEs amidst other entities facilitating data exchange and broader frameworks for data exchange (for example, vendor health information networks, TEFCA, private exchange networks, etc.)?

HIEs can be and often are statewide on-ramps to QHINS.

PC-12. What are the most valuable operational health data use cases for patients and caregivers that, if addressed, would create more efficient care navigation or eliminate barriers to competition among providers or both?

a. Examples may include the following:

(1) Binding cost estimates for pre-defined periods.

(2) Viewing provider schedule availability.

(3) Using third-party apps for appointment management.

(4) Accessing patient-facing quality metrics.

(5) Finding the right provider for specific healthcare needs.

I love the above. As I also mentioned, be able to track costs by condition or medication.

b. What use cases are possible today?
I believe that tracking costs by condition is possible or nearly so today.

c. What should be possible in the near future?
Provider scheduling standards need work, but should be feasible in the near future.

d. What would be very valuable but may be very hard to achieve?

3. Information Blocking and Digital Identity

PC-13. How can CMS encourage patients and caregivers to submit information blocking complaints to ASTP/ONC's Information Blocking Portal? What would be the impact? Would increasing reporting of complaints advance or negatively impact data exchange?

- Maybe require patient portals to provide the capability to easily report concerns to a third party or Federal agency.
- Provide a way to e-mail the complaint directly using a well-known account (e.g., blocking@astp.gov, or better yet, blocking@cms.gov or blocking@hhs.gov. Most everyone knows how to e-mail, few know to find a portal.

PC-14. Regarding digital identity credentials (for example, CLEAR, Login.gov, ID.me, other NIST 800-63-3 IAL2/AAL2 credentialing service providers (CSP)):

a. What are the challenges today in getting patients/caregivers to sign up and use digital identity credentials?

Make it easy to do through a simple form that can be taken to a notary or a local government office (e.g., town clerk) and mailed to a provider of such an identity.

b. What could be the benefits to patients/caregivers if digital identity credentials were more widely used?

Maybe I could have one identity.

c. What are the potential downsides?

Broader use of digital identity services will create a larger attack surface. Expect it, and don't view it as a failure when it happens. This will correct itself over time. Ensure that there's a way to get rid of bad CSPs.

d. How would encouraging the use of CSPs improve access to health information?
I would love to have one CSP login. I have TSA Pre, a government PIV card, a passport, and my state driver's license. Any and all of those could be linked to a digital identity I should be able to use for other purposes. If they are good to get me into a federal building there's NO reason I should have to get another one to log in to my patient portal.

e. What role should CMS/payers, providers, and app developers have in driving adoption?

f. How can CMS encourage patients to get digital identity credentials?
By linking them to other credentials they already have, and funding states and Federal digital identity providers to make them available to users. Also, by enabling payers to pay for such services.

C. Providers

This section is intended for all stakeholders to provide input on questions as they relate to use cases and workflows that involve providers. While we certainly want providers to answer questions in this section (and in other sections) from the provider point of view, we also invite all stakeholders to provide their viewpoints on the provider workflows as appropriate.

1. Digital Health Apps

PR-1. What can CMS and its partners do to encourage providers, including those in rural areas, to leverage approved (see description in PC-5) digital health products for their patients?

a. What are the current obstacles?

b. What information should providers share with patients when using digital products in the provision of their care?

c. What responsibilities do providers have when recommending use of a digital product by a patient?

PR-2. What are obstacles that prevent development, deployment, or effective utilization of the most useful and innovative applications for physician workflows, such as quality measurement reporting, clinical documentation, and billing tasks? How could these obstacles be mitigated?

PR-3. How important is it for healthcare delivery and interoperability in urban and rural areas that all data in an EHR system be accessible for exchange, regardless of storage format (for example, scanned documents, faxed records, lab results, free text notes, structured data fields)? Please address all of the following:

a. Current challenges in accessing different data formats.

b. Impact on patient care quality.

c. Technical barriers to full data accessibility.

d. Cost or privacy implications of making all data formats interoperable.

e. Priority level compared to other interoperability needs.

PR-4. What changes or improvements to standards or policies might be needed for patients' third-party digital products to have access to administrative workflows, such as auto-populating intake forms, viewing provider information and schedules, and making and modifying an appointment?

Enforcing a standard for scheduling via g(10) would be a huge improvement to start with.

Different providers have different workflows and concerns about patient data necessary on intake. What a cardiology, your dentist, primary care, GI specialist and surgeon want to know will vary based on specialty. The FHIR Questionnaire and QuestionnaireResponse resource might be helpful here … and if so, would benefit from a standardized set of inquiry questions and response elements to cover common cases (e.g., to enumerate allergies, conditions, family history, or medications).

2. Data Exchange

PR-5. Which of the following FHIR APIs and capabilities do you already support or utilize in your provider organization's systems, directly or through an intermediary? For each, describe the transaction model, use case, whether you use individual queries or bulk transactions, and any constraints:

a. Patient Access API.

Yes

b. Standardized API for Patient and Population Services.
Yes

c. Provider Directory API.
Yes

d. Provider Access API.
Yes

e. Payer-to-Payer API.

f. Prior Authorization API.

g. Bulk FHIR—Do you support Group ID-based access filtering for population-specific queries?
I'd like to.

h. SMART on FHIR—Do you support both EHR-launched and standalone app access? What does the process for application deployment entail?
Yes

i. CDS Hooks (for clinical decision support integrations).

PR-6. Is TEFCA currently helping to advance provider access to health information?

a. Please provide specific examples.
Yes

b. What changes would you suggest?
Do what you can to accelerate adoption.

c. What other options are available outside of TEFCA?
The eHealthExchange, CommonWell, Carequality nexus of connected networks is OK, but not as valuable as TEFCA, and for EHR developers, building an application to connect via any one network is not a helpful experience in connecting to another. TEFCA should improve this.

d. Are there redundant standards, protocols or channels or both that could be consolidated?

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PR-7. What strategies can CMS implement to support providers in making high-quality, timely, and comprehensive healthcare data available for interoperability in the digital product ecosystem? How can the burden of increasing data availability and sharing be mitigated for providers? Are there ways that workflows or metrics that providers are already motivated to optimize for that could be reused for, or combined with, efforts needed to support interoperability?

PR-8. What are ways CMS or partners can help with simplifying clinical quality data responsibilities of providers?

a. What would be the benefits and downsides of using Bulk FHIR data exports from EHRs to CMS to simplify clinical quality data submissions? Can CMS reduce the burden on providers by performing quality metrics calculations leveraging Bulk FHIR data exports?

b. In what ways can the interoperability and quality reporting responsibilities of providers be consolidated so investments can be dually purposed?

c. Are there requirements CMS should consider for data registries to support digital quality measurement in a more efficient manner? Are there requirements CMS should consider for data registries that would support access to real-time quality data for healthcare providers to inform clinical care in addition to simplifying reporting processes?

3. Digital Identity

PR-9. How might CMS encourage providers to accept digital identity credentials (for example, CLEAR, ID.me, Login.gov) from patients and their partners instead of proprietary logins that need to be tracked for each provider relationship?

a. What would providers need help with to accelerate the transition to a single set of trusted digital identity credentials for the patient to keep track of, instead of one for each provider?

b. How might CMS balance patient privacy with convenience and access to digital health products and services that may lead to significant improvements in health?

PR-10. Regarding digital identity credentials (for example, CLEAR, Login.gov, ID.me, other NIST 800-63-3 IAL2/AAL2 CSPs):

a. What are the challenges and benefits for providers?
The challenges for integrating digital identities for providers is a lack of integration on the EHR to accept a digital identity, and for patients to connect a digital identity to a provider EHR system. Digital identities which also provide a physical identity artifact, or a physical identity card that are linked with digital identities would be much easier to integrate into provider workflows. Consider how using a driver's license, Global Entry (TSA), passport, or CLEAR identity card which has bar-code readable content might simplify integration of digital identity credentials with a provider EHR system and kiosk or other electronic workflows.

b. How would requiring their use improve access to health information?

Requiring their use by patient's will NOT improve access, it will only prevent access by patients who do not have them, cannot easily get them or have other challenges (such as a lack of a permanent address), that might otherwise prevent their use. However, requiring that they be ABLE to be used by healthcare provider systems to verify patient identities could improve

access. Any requirement on providers should include stipulations that there should be other mechanisms to enable patients to get access.

c. What are the potential downsides?

Digital identities suffer from the same downsides as any other identity. They can be stolen, they encourage the accumulation of digital identity data in systems that may not be secure, et cetera.

d. What impact would mandatory credentials have on a nationwide provider directory?
Mandatory digital identity credentials required of licensed healthcare providers might be beneficial to a nationwide provider directory, and could improve data quality in such a directory.

e. How could digital identity implementation improve provider data flow?

f. Would combining FHIR addresses and identity improve data flow?

Identity matching today already has challenges due to address formatting challenges. Much of this is already discussed and addressed in both US@ efforts, and in the FAST Digital Identity and Patient Matching guide.

PR-11. How could members of trust communities [3] (for example, QHINs, participants and subparticipants in TEFCA, which requires Identity Assurance Level 2 (IAL2) via Credential Service Providers (CSPs)) better support the goals of reduced provider and patient burden while also enhancing identity management and security?

Enable better use of IAL2 validated identities across systems, so that no single CSP need to be selected within a trust community. Enable "Bring your own identity" to this space, and provide a consumer option for it that enables consumers to readily obtain a digital identity that they can trust.

4. Information Blocking

PR-12. Should ASTP/ONC consider removing or revising any of the information blocking exceptions or conditions within the exceptions (45 CFR part 171, subparts B through D) to further the access, exchange, and use of electronic health information (EHI) and to promote market competition?

While such consideration may be warranted, I would advise to be careful of responding to requests that may be motivated by special interests, or which heavily favor commercial vendors.

PR-13. For any category of healthcare provider (as defined in 42 U.S.C. 300jj(3)), without a current information blocking disincentive established by CMS, what would be the most effective disincentive for that category of provider?

PR-14. How can CMS encourage providers to submit information blocking complaints to ASTP/ONC's Information Blocking Portal? What would be the impact? Would it advance or negatively impact data exchange?

Portals are NOT user-friendly ways to work with consumers or providers, especially as a way to first engage with an agency such as CMS or ASTP/ONC. As a user, the easiest thing for me to do is to write an e-mail to a well-known address (e.g., blocking@cms.gov). Use AI , e-mail and the web follow up on the issue to break it down into specifics that you need, e.g., provider, technology in use, et cetera.

D. Payers

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PA-1. What policy or technical limitations do you see in TEFCA? What changes would you suggest to address those limitations? To what degree do you expect these limitations to hinder participation in TEFCA?

PA-2. How can CMS encourage payers to accelerate the implementation and utilization of APIs for patients, providers, and other payers, similar to the Blue Button 2.0 and Data at the Point of Care APIs released by CMS?

PA-3. How can CMS encourage payers to accept digital identity credentials (for example, CLEAR, ID.me, Login.gov) from patients and their partners instead of proprietary logins?

PA-4. What would be the value to payers of a nationwide provider directory that included FHIR end points and used digital identity credentials?

PA-5. What are ways payers can help with simplifying clinical quality data responsibilities of providers?

a. How interested are payers and providers in EHR technology advances that enable bulk extraction of clinical quality data from EHRs to payers to allow them to do the calculations instead of the provider-side technology?

b. In what ways can the interoperability and quality reporting responsibilities of providers to both CMS and other payers be consolidated so investments can be dually purposed? Are there technologies payers might leverage that would support access to real time quality data for healthcare providers to inform clinical care in addition to simplifying reporting processes?

PA-7. How can CMS encourage payers to submit information blocking complaints to ASTP/ONC's Information Blocking Portal? What would be the impact? Would it advance or negatively impact data exchange?

E. Technology Vendors, Data Providers, and Networks

This section is intended for all stakeholders to provide input on questions as they relate to use cases and workflows that involve technology vendors, data providers, and networks. While we certainly want technology vendors, data providers, and networks to answer questions in this section (and in other sections) from their point of view, we also invite all stakeholders to provide their viewpoints on the technology vendor, data provider, and network use cases as appropriate.

1. Ecosystem

TD-1. What short term (in the next 2 years) and longer-term steps can CMS take to stimulate developer interest in building digital health products for Medicare beneficiaries and caregivers?

TD-2. Regarding CMS Data, to stimulate developer interest—

a. What additional data would be most valuable if made available through CMS APIs?

b. What data sources are most valuable alongside the data available through the Blue Button 2.0 API?

c. What obstacles prevent accessing these data sources today?

d. What other APIs should CMS and ASTP/ONC consider including in program policies to unleash innovation and support patients and providers?

2. Digital Identity

TD-3. Regarding digital identity implementation:

a. What are the challenges and benefits?

b. How would requiring digital identity credentials (for example, CLEAR, Login.gov, ID.me, other NIST 800-63-3 IAL2/AAL2 CSPs) impact cybersecurity and data exchange?

c. What impact would mandatory use of the OpenID Connect identity protocol have?

3. Technical Standards and Certification

TD-4. How can CMS better encourage use of open, standards-based, publicly available APIs over proprietary APIs?

TD-5. How could a nationwide provider directory of FHIR endpoints improve access to health information for patients, providers, and payers? Who should publish such a directory, and should users bear a cost?

TD-6. What unique interoperability functions does TEFCA perform?

a. What existing alternatives should be considered?

b. Are there redundant standards, protocols or channels or both that should be consolidated?

TD-7. To what degree has USCDI improved interoperability and exchange and what are its limitations?

a. Does it contain the full extent of data elements you need?

b. If not, is it because of limitations in the definition of the USCDI format or the way it is utilized?

c. If so, would adding more data elements to USCDI add value or create scoping challenges? How could such challenges be addressed?

d. Given improvements in language models, would you prefer a non-proprietary but less structured format that might improve data coverage even if it requires more processing by the receiver?

TD-8. What are the most effective certification criteria and standards under the ONC Health IT Certification Program?

TD-9. Regarding certification of health IT:

a. What are the benefits of redefining certification to prioritize API-enabled capabilities over software functionality?

b. What would be the drawbacks?

c. How could ASTP/ONC revise health IT certification criteria to require APIs to consistently support exchanging data from all aspects of the patient's chart (for example, faxed records, free text, discrete data)?

d. What policy changes could CMS make so providers are motivated to respond to API-based data requests with best possible coverage and quality of data?

e. How could EHRs capable of bulk data transfer be used to reduce the burden on providers for reporting quality performance data to CMS? What capabilities are needed to show benefit? What concerns are there with this approach?

TD-10. For EHR and other developers subject to the ONC Health IT Certification Program, what further steps should ASTP/ONC consider to implement the 21st Century Cures Act's API condition of certification (42 U.S.C. 300jj-11(c)(5)(D)(iv)) that requires a developer's APIs to allow health information to be accessed, exchanged, and used without special effort, including providing access to all data elements of a patient's electronic health record to the extent permissible under applicable privacy laws?

TD-11. As of January 1, 2024, many health IT developers with products certified through the ONC Health IT Certification Program are required to include the capability to perform an electronic health information export or “EHI export” for a single patient as well as for patient

populations (45 CFR 170.315(b)(10)). Such health IT developers are also required to publicly describe the format of the EHI export. Notably, how EHI export was accomplished was left entirely to the health IT developer. Now that this capability has been in production for over a year, CMS and ASTP/ONC seek input on the following:

a. Should this capability be revised to specify standardized API requirements for EHI export?

b. Are there specific workflow aspects that could be improved?

c. Should CMS consider policy changes to support this capability's use?

4. Data Exchange

TD-12. Should CMS endorse non-CMS data sources and networks, and if so, what criteria or metrics should CMS consider?

CMS should get onto TEFCA.

TD-13. What new opportunities and advancements could emerge with APIs providing access to the entirety of a patient's electronic health information (EHI)?

a. What are the primary obstacles to this?

b. What are the primary tradeoffs between USCDI and full EHI, especially given more flexible data processing capabilities today?

Full EHI is junk. That's the kindest four-letter word I can use to describe it. USCDI is standardized and understandable outside of a proprietary context.

TD-14. Regarding networks' use of FHIR APIs:

a. How many endpoints is your network connected to for patient data sharing? What types, categories, geographies of endpoints do you cover? Are they searchable by National Provider Identifier (NPI) or organizational ID?

My employers network is connected to 47 states, 27,000+ LTPAC facilities, 2800+ hospitals, 2000+ ambulatory clinics, and every major US health plan.

b. How are these connections established (for example, FHIR (g)(10) endpoints, TEFCA/Integrating the Health Enterprise (IHE) XCA, or proprietary APIs)?

All of the above.

c. Do you interconnect with other networks? Under what frameworks (for example, TEFCA, private agreements)?
TEFCA

TD-15. Regarding bulk FHIR APIs:

a. How would increased use of bulk FHIR improve use cases and data flow?
Bulk FHIR APIs would enable public health, schools, employers, payers, care management organizations, and LTPAC facilities to better collect data on the individuals for which they may need healthcare data (with consent of the patient in some patients).

b. What are the potential disadvantages of their use?

Again, any bulk operation is going to increase attack surface, and provide a source of risk. This should not be weighed heavily against the use of it, as over time, these risks will be mitigated.

TD-16. What are the tradeoffs of maintaining point-to-point models vs. shared network infrastructure?

a. Do current rules encourage scalable network participation?

b. What changes would improve alignment (for example, API unification, reciprocal access)?

TD-17. Given operational costs, what role should CMS or ASTP/ONC or both have in ensuring viability of healthcare data sharing networks, including enough supply and demand, that results in usage and outcomes?

Both CMS and ASTP both have a role in ensuring viability of healthcare data sharing networks. ASTP's role is in ensuring that these networks enforce common standards with certified health IT products. CMS role is to incentivize the use of these networks by the healthcare providers and payers that it has relationships with. CMS should also ensure that it takes part in the use of TEFCA, likely by engaging with eHealth Exchange as that is the de-facto QHIN used by every other government agency.

5. Compliance

TD-18. Information blocking:

a. Could you, as a technology vendor, provide examples for the types of practices you have experienced that may constitute information blocking. Please include both situations of non-responsiveness as well as situations that may cause a failure or unusable response?

b. What additional policies could ASTP/ONC and CMS implement to further discourage healthcare providers from engaging in information blocking practices?

c. Are there specific categories of healthcare actors covered under the definition of information blocking in section 3022(a)(1) of the Public Health Service Act (PHSA) that lack information blocking disincentives?

TD-19. Regarding price transparency implementation:

a. What are current shortcomings in content, format, delivery, and timeliness?

Individual price transparency may be available, but realistically, not in a format that is useable by many. More work is clearly needed in this area.

b. Which workflows would benefit most from functional price transparency?

c. What improvements would be most valuable for patients, providers, or payers, including CMS?

Enabling patients and their caregivers to take better control of their healthcare costs through price transparency should be a huge workflow improvement for those individuals, and could potentially enable significant cost savings for CMS. However, in order for individuals (and others) to use price transparency data, there should be a standardized format for it, and rather than self-publishing it in hard to find web URLs, providers should annually submit price transparency data to a neutral third party that can make it more usable and available for others.

d. What would further motivate solution development?

F. Value-Based Care Organizations

This section is intended for all stakeholders to provide input on questions as they relate to use cases and workflows that involve value-based care organizations. While we certainly want value-based care organizations to answer questions in this section (and in other sections) from the value-based care provider point of view, we also invite all stakeholders to provide their viewpoints on the value-based care workflows as appropriate.

1. Digital Health Adoption

VB-1. What incentives could encourage APMs such as accountable care organizations (ACOs) or participants in Medicare Shared Savings Program (MSSP) to leverage digital health management and care navigation products more often and more effectively with their patients? What are the current obstacles preventing broader digital product adoption for patients in ACOs?

VB-2. How can key themes and technologies such as artificial intelligence, population health analytics, risk stratification, care coordination, usability, quality measurement, and patient engagement be better integrated into APM requirements?

VB-3. What are essential health IT capabilities for value-based care arrangements?

a. Examples (not comprehensive) may include: care planning, patient event notification, data extraction/normalization, quality performance measurement, access to claims data, attribution and patient ID matching, remote device interoperability, or other patient empowerment tools.

b. What other health IT capabilities have proven valuable to succeeding in value-based care arrangements?

VB-4. What are the essential data types needed for successful participation in value-based care arrangements?

2. Compliance and Certification

VB-5. In your experience, how do current certification criteria and standards incorporated into the ONC Health IT Certification Program support value-based care delivery?

VB-6. What specific health information technology capabilities that could benefit APMs are not currently addressed by existing certification criteria and standards that should be included under the ONC Health IT Certification Program?

VB-7. How can technology requirements for APMs, established through CEHRT or other pathways, reduce complexity while preserving necessary flexibility?

VB-8. How can other HHS policies supplement CEHRT requirements to better optimize the use of digital health products in APMs? As an example, requirements under the Conditions of Participation for hospitals (42 CFR 482.24(d)) require hospitals to transmit electronic patient event notifications to community providers. What barriers are in place preventing APM participants from receiving the same notifications?

VB-9. What technology requirements should be different for APM organizations when comparing to non-APM organizations (for example, quality reporting, and interoperability)?

VB-10. In the Calendar Year (CY) 2024 Physician Fee Schedule final rule (88 FR 79413), CMS established that CEHRT requirements for Advanced APMs beyond those in the “Base EHR” definition should be flexible based on what is applicable to the APM that year based on the area of clinical practice. What certification criteria should CMS identify under this flexibility for specific Advanced APMs, or for Advanced APMs in general? Are there specific flexibilities or alternatives to consider for smaller or resource-constrained (such as rural) providers in meeting CEHRT requirements without compromising quality of care or availability of performance data?

3. Technical Standards

VB-11. What specific interoperability challenges have you encountered in implementing value-based care programs?

VB-12. What technology standardization would preserve program-specific flexibility while promoting innovation in APM technology implementation?

VB-13. What improvements to existing criteria and standards would better support value-based care capabilities while reducing provider burden?

VB-14. How could implementing digital identity credentials improve value-based care delivery and outcomes?

VB-15. How could a nationwide provider directory of FHIR endpoints help improve access to patient data and understanding of claims data sources? What key data elements would be necessary in a nationwide FHIR endpoints directory to maximize its effectiveness?